KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

DIGITAL STUDENT ID AND PROFILE MANAGEMENT

Submitted by

Arul kumaran k S

25BIT010

Under the Guidance of

DHILEEPAN A

Technical Trainer

Technical Department, KG College of Arts and Science

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Semester: **I**

DECLARATION

I hereby declare that the project entitled "**DIGITAL STUDENT ID AND PROFILE MANAGEMENT"** submitted in partial fulfillment of the requirements for the course [**JAVA PROGRAMMING]** is a **record of my original work** carried out under the guidance of **[Dhuleepan a]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Arul kumaran KS

CERTIFICATE

This is to certify that the project entitled **"DIGITAL STUDENT ID AND PROFILE MANAGEMENT"** is a **bona fide work** carried out by **[Arul Kumaran KS] [I-B.SC. INFORMATION TECHNOLOGY]**, [**INFORMATION TECHNOLOGY]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09-04-2026

Submitted for the Viva-Voce Examination held on 09-04-2026

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Digital Student ID and Profile Management System is a Java-based application developed using Java Swing GUI to manage student information digitally. In many educational institutions, student details and ID information are maintained manually, which can lead to data loss, errors, and difficulty in managing records.

This project provides a computerized system where student information such as Student ID, Name, Department, Batch, and Blood Group can be stored and displayed efficiently. The system allows the user to add student details and view them in a structured format.

The application is developed using Core Java concepts such as classes, objects, arrays, collections, and GUI components. Java Swing is used to create a user-friendly graphical interface that allows easy interaction with the system.

The main goal of this system is to simplify student data management, reduce manual record keeping, and provide a quick way to generate and view student ID information digitally.

1. INTRODUCTION

In educational institutions, managing student information is an important task. Traditionally, student records and ID cards are handled manually, which can create several issues such as data errors, difficulty in retrieving records, and time-consuming management.

With the help of computer technology, student information can be stored and managed digitally. A digital system helps institutions organize student data efficiently and access it quickly whenever required.

The Digital Student ID and Profile Management System is designed to store and manage student information using Java programming. The system allows users to enter student details and generate a digital student ID profile.

The project is implemented using Java Swing GUI, which provides a graphical interface for entering and displaying student data.

This project demonstrates the practical use of Object-Oriented Programming (OOP) concepts in Java and shows how Java applications can be used to manage real-world data efficiently.

1.1 OBJECTIVE

The main objective of the Digital Student ID and Profile Management System is to develop a Java-based application that can store and manage student information digitally.

The specific objectives include:

To develop a Java Swing GUI application for managing student data.

To store student details such as ID, Name, Department, Batch, and Blood Group.

To display student profiles in a structured format.

To reduce manual record management.

To demonstrate the implementation of Java OOP concepts.

To create a simple and user-friendly interface for managing student information.

1.2 OVERVIEW

The Digital Student ID and Profile Management System is a desktop-based application developed using Java Swing.

The system allows the user to:

Enter student information

Store student details

Display student profiles

Manage multiple student records

The application provides a graphical interface with text fields and buttons where the user can input the student details. Once the information is entered, the system stores the details and displays them in an organized format.

The system ensures that certain fields follow predefined conditions such as:

Blood Group: O+, O-, A+, A-, B+, B-, AB+, AB-

Batch: 2024, 2025, 2026

Department: BSc IT, BSc CS, BCA, BCom, etc.

This helps maintain data consistency and accuracy.

1.3 FEATURES

The Digital Student ID System provides several important features:

1. Student Data Entry

Allows the user to enter student details such as ID, Name, Department, Batch, and Blood Group.

2. Profile Management

Stores and manages student profiles efficiently.

3. Data Validation

Ensures that fields like blood group, department, and batch follow specific values.

4. GUI Interface

Provides a user-friendly graphical interface using Java Swing.

5. Organized Data Display

Displays student information in a structured format.

6. Digital Record Management

1. Maintains student records digitally instead of manual files.

2. PROGRAM ANALYSIS

2.1 Problem Definition

In many colleges, student information is stored manually using paper records or simple spreadsheets. This approach has several limitations.

Common problems include:

Difficulty in maintaining student records

Risk of losing important information

Time-consuming manual updates

Lack of centralized data storage

Difficulty in searching student records

To solve these issues, a digital system is required that can manage student information efficiently.

The Digital Student ID and Profile Management System provides a computerized solution for storing and managing student details.

2.2 Project Overview

The system allows users to input student details through a graphical user interface.

The main functionalities include:

Adding student information

Displaying student profiles

Organizing student data

Managing multiple student records

The system uses Java Swing components such as:

JFrame

JLabel

JTextField

JComboBox

JButton

JTextArea

The data entered by the user is stored using ArrayList collections in Java.

2.3 Scope and Limitations

Scope

The system can be used in:

Colleges

Universities

Educational institutions

Training centers

The system helps manage student ID information digitally.

Limitations

Some limitations of the current system include:

No database integration

Data is stored only during program execution

Limited security features

Future versions of the system can include database connectivity and login authentication.

3. SYSTEM DESIGN

3.1 Modules

1. Login Module

Allows authorized users to access the system.

2. Student Information Module

Collects student details such as:

Student ID

Name

Department

Batch

Blood Group

3. Data Validation Module

Ensures that input values follow predefined rules.

Example:

Blood Group → O+, O-, A+, A-, B+, B-, AB+, AB-

Batch → 2024, 2025, 2026

4. METHODOLOGY

Displays the student profile in a structured format.

5. SOURCE CODE

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
```

```
public class ScjolarshipEligiblity {

    public static void main(String[] args) {

                        // Frame
                        JFrame f = new JFrame("Scholarship Eligibility");
                        f.setVisible(true);
                        f.setSize(400, 300);
                        f.setLayout(null);
                        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

                        // Labels
                        JLabel l1 = new JLabel("Marks:");
                        JLabel l2 = new JLabel("Income:");
                        JLabel result = new JLabel("");

                        // TextFields
                        JTextField t1 = new JTextField();
                        JTextField t2 = new JTextField();

                        // Button
                        JButton b = new JButton("Check");

                        // Set Positions
                        l1.setBounds(50, 50, 100, 30);
                        t1.setBounds(150, 50, 150, 30);

                        l2.setBounds(50, 100, 100, 30);
                        t2.setBounds(150, 100, 150, 30);
```

```
b.setBounds(120, 150, 100, 30);

result.setBounds(50, 200, 300, 30);

// Button Action
b.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        try {
            int marks = Integer.parseInt(t1.getText());
            int income = Integer.parseInt(t2.getText());

            // Eligibility Condition
            if (marks >= 75 && income <= 150000) {
                result.setText("Eligible for Scholarship ");
            } else {
                result.setText("Not Eligible ");
            }

        } catch (Exception ex) {
            result.setText("Enter valid numbers!");
        }
    }
});

// Add Components
f.add(l1);
f.add(t1);
f.add(l2);
```

```
                f.add(t2);
                f.add(b);
                f.add(result);
        }

}
```

6. INPUT AND OUTPUT SCREENS

Include screenshots such as:

Figure 1: Login Screen

Figure 2: Student Entry Form

Figure 3: Student Profile Output

Figure 4: Student List Display

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Digital Student ID and Profile Management System demonstrates how Java can be used to build simple desktop applications for managing student information.

The system successfully stores student data and displays profiles using a graphical interface.

7.2 Conclusion

The project successfully developed a Java Swing GUI application for managing student ID profiles digitally.

The system reduces manual work and provides an organized way to manage student information.

Future improvements may include:

Database integration

Search functionality

Student photo upload

Admin login system

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com